

Advanced Calculus

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Below, we collect some useful results from advanced calculus. (This was intended to make the technical details of Fourier analysis more transparent.)

Lebesgue criterion:

Recall that the standard domain of Riemann integration is a closed cell $\Delta \subset \mathbb{R}^n$.

Theorem 0.1 (Lebesgue criterion)

f is integrable on $\Delta \subset \mathbb{R}^n$ iff it is discontinuous on a set of measure 0 (i.e. it is continuous a.e.).

Fubini's theorem:

Source: Gunning, Chapter 6-2

Theorem 0.2

Suppose $f(x_1, x_2)$ is integrable on $\Delta = \Delta_1 \times \Delta_2 \subset \mathbb{R}^n$, where $\Delta_1 \subset \mathbb{R}^{n_1}$, $\Delta_2 \subset \mathbb{R}^{n_2}$. Assume that $f(x_1, x_2)$ is integrable with respect to x_1 . Then $\int_{x_1 \in \Delta_1} f(x_1, x_2) dx_1$ is integrable with respect to x_2 and we have the equality

$$\int_{\Delta} f dx = \int_{x_2 \in \Delta_2} \int_{x_1 \in \Delta_1} f(x_1, x_2) dx_1 dx_2$$

Corollary 0.3

If $f(x_1, x_2)$ is continuous on $\Delta = \Delta_1 \times \Delta_2 \subset \mathbb{R}^n$, then

$$\int_{\Delta} f dx = \int_{x_1 \in \Delta_2} \int_{x_1 \in \Delta_1} f(x_1, x_2) dx_1 dx_2 = \int_{x_1 \in \Delta_1} \int_{x_2 \in \Delta_2} f(x_1, x_2) dx_2 dx_1$$

Remark 0.4. Note that if f is integrable on Δ , it *does not* follow that for fixed x_2 , $f(x_1, x_2)$ is integrable with respect to x_1 . This is why we assume that $f(x_1, x_2)$ is integrable with respect to x_1 .

Remark 0.5. Practically, when using Fubini for a function f that is simply integrable, be sure to check that $f(x_1, x_2)$ is integrable with respect to x_1 . When f is continuous, the corollary says that this is not necessary.

Add Gunning definition of integral on Jordan measurable set...

Add COV formula (assuming Jordan measurable...)

Add spherical coordinates...

Add divergence theorem (and how differentials change)...

Improper integrals:

Add general limiting definition from Gunning...

Add discussion of moderate decrease from Stein-Shakarchi...

Add statement of COV...

Add spherical coordinates...

Interchange theorems:

Source: Lang, "Undergraduate Analysis", Chapters 9-2 + 13-3

Limits/integrals/derivatives of sequences:

Theorem 0.6

If $\{f_n\}$ are continuous on $[a, b]$ and $\{f_n\}$ converges uniformly to f on $[a, b]$, then $\lim_{x \rightarrow x_0} f_n(x) = f(x_0)$.

Proof. Clear from the fact that the uniform limit f must be continuous. $\square$

Theorem 0.7

If $\{f_n\}$ are continuous on $[a, b]$ and $\{f_n\}$ converges uniformly to f on $[a, b]$, then $\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b f$.

Theorem 0.8

Suppose that $\{f_n\}$ are C^1 on $[a, b]$, $\{f'_n\}$ converges uniformly to a function g , and there exists $x_0 \in [a, b]$ for which $\{f_n(x_0)\}$ converges. Then $\{f_n\}$ converges uniformly to a differentiable function f such that $f' = g$.

Limits/derivatives of series:

These results are corollaries of the results on sequences, as a series is simply the limit of its partial sums.

Theorem 0.9

If $\{f_n\}$ are continuous on $[a, b]$ and the series $\sum f_n$ converges uniformly to f , then $\lim_{x \rightarrow x_0} \sum f_n(x) = f(x_0)$.

Theorem 0.10

Suppose that f_n are C^1 on $[a, b]$, the series $\sum f'_n$ converges uniformly to a function g , and there exists x_0 for which $\sum f_n(x_0)$ converges. Then $\sum f_n$ converges uniformly to a differentiable function f such that $f' = \sum_n f'_n$.

Limits of (improper) integrals:

Definition 0.11. Let $\phi(y) = \int_a^\infty f(x, y)dx$, where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is integrable. We say that $\phi(y)$ **converges uniformly** in y if the limit $\lim_{b \rightarrow \infty} \int_a^b f(x, y)dx$ converges uniformly in y .

Remark 0.12. Note that the definition may be extended to all other cases when a or b is infinite.

Theorem 0.13

Let $\phi(y) = \int_a^b f(x, y)dx$, where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is integrable. (Here, a, b are allowed to be $\pm\infty$.) Suppose that f is continuous on $[a, b] \times \mathbb{R}$ and $\int_a^b f(x, y)dx$ converges uniformly in y . Then for all $y \in \mathbb{R}$, $\lim_{y \rightarrow y_0} \phi(y) = \int_a^b f(x, y_0)dx$, i.e. $\phi(y)$ is continuous on $\mathbb{R}$.

Proof. Sketch: write out the difference. When the integral is improper, split the integral into a bounded interval and tails. Use uniform continuity for the interval and uniform convergence for the tails. $\square$

Remark 0.14. If a, b are finite, we do not need uniform convergence. It suffices to check that f is continuous on $(x, y) \in [a, b] \times \mathbb{R}$. Further, if we only need to show the result for some $y_0 \in \mathbb{R}$, it is sufficient that f is continuous on a compact neighborhood of y_0 .

Derivatives of (improper) integrals (Leibniz rule):

Sources: Lang (above) and Conrad, <https://kconrad.math.uconn.edu/blurbs/analysis/diffunderint.pdf>

Theorem 0.15 (Leibniz rule)

Let $\phi(y) = \int_a^b f(x, y)dx$, where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is integrable. (Here, a, b are allowed to be $\pm\infty$.) Suppose that:

1. f and $\partial_y f$ are continuous on $[a, b] \times \mathbb{R}$;
2. there exist $A(x), B(x)$ s.t. $|f(x, y)| \leq A(x)$, $|\partial_y f(x, y)| \leq B(x)$ with $\int_a^b A(x)$ and $\int_a^b B(x)$ convergent.

Then $\phi(y)$ is differentiable and $\partial_y \phi(y) = \int_a^b \partial_y f(x, y)dx$.

Proof. Sketch: write out the difference quotient. When the integral is improper, split the integral into a bounded interval and tails. Use 1 for the interval and 2 for the tails. $\square$

Remark 0.16. If a, b are finite, we do not need condition 1. It suffices to check that $f, \partial_y f$ are continuous on $(x, y) \in [a, b] \times \mathbb{R}$. Further, if we only need to show the result for some $y_0 \in \mathbb{R}$, it is sufficient that $f, \partial_y f$ are continuous on a compact neighborhood of y_0 .

Remark 0.17. Condition 2 may be regarded as “uniform convergence” of $\int f, \int \partial_y f$ in the variable y . In this light, we see the connection to interchange of derivatives and series of functions, for which uniform convergence is also required. (Note: Lang specifically refers to condition 2 as “absolute convergence”.)